



DEPARTMENT OF MATHEMATICS
NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA

B.Tech. Second Semester (EE/CE/ECE/ME/RA)

Mid Term – II Examination

Date: April 07, 2025

Max. Marks: 15

Time: 11:00 am – 11:50 am

Paper: Integral Calculus and Difference Equations (MAIC 102)

Instructions:

- Answer all the questions.
- The calculator is not allowed into the examination hall.
- Start writing your answers from the first page. Do not leave it blank. No extra answer sheets will be provided.

1. Use spherical co-ordinates to evaluate the integral $\iiint \frac{dx dy dz}{\sqrt{x^2 + y^2 + z^2}}$ taken over the region V in the first octant bounded by the cones $\theta = \frac{\pi}{4}$ and $\theta = \tan^{-1}(2)$ and the sphere $x^2 + y^2 + z^2 = 6$. [5M]
2. Find $f(r)$ such that $\nabla f = \frac{\vec{r}}{r^5}$ and $f(1) = 0$. [5M]
3. Evaluate the surface integral $\iint \vec{F} \cdot \hat{n} ds$ over S , where $\vec{F} = 6z\hat{i} + 6\hat{j} + 3y\hat{k}$ and S is the portion the plane $2x + 3y + 4z = 12$ which in the first octant. [5M]